NOTES FOR TMTS

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Contents

1 Useful Conclusions and Fundamental Definitions **3**

1.1 Probability Theory 3

1.2 Definitions 3

2 Stationary ARMA Processes **6**

2.1 ARMA Processes 6

3 Stationary ARMA Processes: Further Theory (Chapter 3) **10**

3.1 Stationary Solutions and Laurent Expansions (Section 3.1) 10

3.2 Infinite Moving Averages (Section 3.2) 10

3.3 Coefficient and Covariance Identities (Section 3.3) 11

3.4 Partial Autocorrelation (Section 3.4) 11

3.5 Autocovariance Generating Functions (Section 3.5) 12

3.6 Linear Difference Equations (Section 3.6) 13

4 The Spectral Representation of a Stationary Process (Chapter 4) **14**

4.1 Complex Stationarity and Spectral Measures (Sections 4.1–4.3) 14

4.2 Filtering, ARMA Spectra and Approximation (Section 4.4) 15

4.3 Covariance Matrices and Fourier Coordinates (Section 4.5) 15

4.4 Orthogonal Increments and Stochastic Integration (Sections 4.6–4.7) 16

4.5 Spectral Representation and Inversion (Sections 4.8–4.9) 17

4.6 General Linear Filters and Fourier Approximation (Sections 4.10–4.11) 18

5 Prediction of Stationary Processes (Chapter 5) **19**

5.1 Prediction Equations and Optimality (Section 5.1) 19

5.2 Recursive Prediction and Innovations (Section 5.2) 19

5.3 Finite-History ARMA Prediction (Section 5.3) 21

5.4 Gaussian Prediction and Prediction Bounds (Section 5.4) 22

5.5 Prediction from the Infinite Past (Section 5.5) 22

5.6 Prediction in the Frequency Domain (Section 5.6) 23

5.7 The Wold Decomposition (Section 5.7) 23

5.8 Kolmogorov's Formula (Section 5.8) 24

A **25**

A.1 25

1 Useful Conclusions and Fundamental Definitions

1.1 Probability Theory

Theorem 1.1.1 (Kolmogorove's Theorem)**.** The probability distribution functions $(F_{\mathbf{t}})_{\mathbf{t}\in\mathcal{T}}$ are the distribution functions of some stochastic process if and only if for any integer n and $\mathbf{t} = (t_1, \cdots, t_n) \in \mathcal{T}$ and $1 \leq i \leq n$,

$$\lim_{x_i \to \infty} F_{\mathbf{t}}(x) = F_{\mathbf{t}(i)}(x(i))$$

where $\mathbf{t}(i)$ and $x(i)$ are the $(n-1)$-component vectors obtained by deleting the i th components of $\mathbf{t}$ and x respectively.

Theorem 1.1.2 (Spectral Theorem for Compact Self-Adjoint Operators)**.** Let H be a real or complex Hilbert space, and let $T : H \to H$ be a compact self-adjoint operator. Then there exists an orthonormal family $\{e_n\}$ consisting of eigenvectors of T corresponding to nonzero eigenvalues $\{\lambda_n\}$ such that

$$H = \ker T \oplus \overline{\operatorname{span}\{e_n\}}.$$

Moreover, the following properties hold:

1. Every eigenvalue λ_n is real.
2. Every nonzero eigenspace is finite-dimensional.
3. The set of nonzero eigenvalues is at most countable, and if there are infinitely many of them, then 0 is their only possible accumulation point.
4. For every $x \in H$,
$$Tx = \sum_n \lambda_n \langle x, e_n \rangle e_n.$$

In particular, if $\ker T = \{0\}$, then $\{e_n\}$ is an orthonormal basis of H.

Proof. See Appendix (A.1.1). □

1.2 Definitions

Definiton 1.2.1 (Stationarity)**.** The time series $(X_t)_{t\in\mathbb{Z}}$ is said to be stationary if

- $\mathbb{E}|X_t|^2 < \infty$ for all $t \in \mathbb{Z}$.
- $\mathbb{E}X_t = m$ for all $t \in \mathbb{Z}$.
- $\operatorname{Cov}(X_s, X_t) =: \gamma_X(s,t) = \gamma_X(s+h, r+h)$ for all $s, t, h \in \mathbb{Z}$.

Definiton 1.2.2 (Strict Stationarity)**.** The time series $(X_t)_{t\in\mathbb{Z}}$ is said to be strictly stationary if the joint distributions of $(X_{t_1}, \cdots, X_{t_k})$ and $(X_{t_1+h}, \cdots, X_{t_k+h})$ are the same for all positive integers k and for all $t_1, \cdots, t_k, h \in \mathbb{Z}$.

The strict stationarity may imply the stationarity easily.

Definiton 1.2.3 (Elementary Properties)**.** . For a stationary process, we define the autocovariance $\gamma_X(h) = \gamma_X(t+h,t)$ for any $t \in \mathbb{Z}$ which is well-defined.

If γ is the autocovariance function of a stationary process $(X_t)_{t\in\mathbb{Z}}$, then

$$\gamma(0) \geq 0, \quad |\gamma(h)| \leq \gamma(0) \text{ for all } h \in \mathbb{Z}$$

and $\gamma(h) = \gamma(-h)$ for any $h \in \mathbb{Z}$.

Proof. We only prove that $\gamma(0) \geq |\gamma(h)|$ for any $h \in \mathbb{Z}$.

$$\begin{aligned}\gamma(h)^2 &= (\mathbb{E}(X_0 - \mu(0))(X_h - \mu(h)))^2 \\ &\leq \mathbb{E}(X_0 - \mu(0))^2 \mathbb{E}(X_h - \mu(h))^2 = \gamma(0)^2\end{aligned}$$

by Cauchy-Schwarz Inequality. □

Definiton 1.2.4 (Non-Negative Definiteness)**.** A real-valued function on the integers, $\kappa : \mathbb{Z} \to \mathbb{R}$ is said to be non-negative definite if and only if

$$\sum_{i,j=1}^{n} a_i \kappa(t_i, t_j) a_j \geq 0$$

for all positive integers n and for all vectors $a = (a_1, \cdots, a_n) \in \mathbb{R}^n$ and $t = (t_1, \cdots, t_n) \in \mathbb{Z}$.

Remark. Replace t_i with i in the definition above will obtain an equivalent conclusion.

Theorem 1.2.1 (Characterization of Autocovariance Functions)**.** A real-valued function defined on the integers is the autocovariance function of a stationary time series if and only if it is even and non-negative definite.

Definiton 1.2.5 (Sample Autocovariance Function)**.** The sample autocovariance function of $\{x_1, \cdots, x_n\}$ is defined by

$$\hat{\gamma}(h) := \frac{1}{n} \sum_{i=1}^{n-h} (x_i - \bar{x})(x_{i+h} - \bar{x}), 0 \leq h < n$$

and $\gamma(h) = \gamma(-h), -n < h \leq 0$.

Proposition 1.2.2. The covariance matrix is symmetric and non-negative definite for any random vector.

Proposition 1.2.3. If $\mathbf{a}$ is an m-component column vector, B is an $m \times n$ matrix, and $\mathbf{X} = (X_1, \cdots, X_n)$ where $\mathbb{E}|X_i|^2 < \infty$, i = $1, \cdots, n$, then the random vector,

$$\mathbf{Y} = \mathbf{a} + B\mathbf{X}$$

has the mean $\mathbb{E}\mathbf{Y} = \mathbf{a} + B\mathbb{E}\mathbf{X}$ and covariance matrix

$$\Sigma_{\mathbf{YY}} = B\Sigma_{\mathbf{XX}}B^T$$

Definiton 1.2.6 (Multivariate Normal Distribution)**.** The random vector $\mathbf{Y} = (Y_1, \cdots, Y_n)$ is said to be multivariate normal, or to have a multivariate normal distribution, if and only if there exist a column vector $\mathbf{a}$, a matrix B and a random vector $\mathbf{X} = (X_1, \cdots, X_m)$ with

independent standard normal components, such that $\mathbf{Y} = \mathbf{a} + B\mathbf{X}$, with

$$\mathbb{E}\mathbf{Y} = \mathbf{a}, \quad \Sigma\mathbf{Y} = BB^T$$

Proposition 1.2.4. If $\mathbf{Y} = (Y_1, \cdots, Y_n)$ is a multivariate normal random vector such that $\mathbb{E}\mathbf{Y} = \mu$ and $\Sigma_{\mathbf{YY}} = \Sigma$, then the characteristic function of $\mathbf{Y}$ is

$$\phi_{\mathbf{Y}}(\mathbf{x}) = \exp\left(i\mathbf{x}^T\mu - \frac{1}{2}\mathbf{x}^T\Sigma\mathbf{x}\right)$$

If $\det\Sigma > 0$ then $\mathbf{Y}$ has the density

$$f_{\mathbf{Y}}(\mathbf{y}) = (2\pi)^{-n/2}(\det\Sigma)^{-1/2}\exp\left[-\frac{1}{2}(\mathbf{y}-\mu)'\Sigma^{-1}(\mathbf{y}-\mu)\right]. \tag{1.6.12}$$

Proof. Consider $\mathbf{X}$ to be an independent standard normal vector, then we know $\mu+\Sigma^{1/2}\mathbf{X} \stackrel{d}{=} \mathbf{Y}$, then

$$\begin{aligned}\phi_{\mathbf{Y}}(\mathbf{x}) &= \mathbb{E}e^{i\mathbf{x}^T\mathbf{Y}} = \mathbb{E}e^{i\mathbf{x}^T(\mu+\Sigma^{1/2}\mathbf{X})} \\ &= \exp(i\mathbf{x}^T\mu)\phi_{\mathbf{X}}(\Sigma^{1/2}\mathbf{x}) = \exp\left(i\mathbf{x}^T\mu - \frac{1}{2}\mathbf{x}^T\Sigma\mathbf{x}\right)\end{aligned}$$

□

2 Stationary ARMA Processes

2.1 ARMA Processes

Definiton 2.1.1 (White Noise)**.** The process (Z_t) is said to be white noise with mean 0 and variance σ^2 if and only if (Z_t) has zero mean and covariance function

$$\gamma(h) = \begin{cases} 0, & h \neq 0 \\ \sigma^2, & h = 0 \end{cases}$$

Definiton 2.1.2 (The ARMA(p,q) Process)**.** The process $(X_t)_{t\in\mathbb{Z}}$ is said to be an ARMA -(p,q) process if (X_t) is stationary and if for every t,

$$X_t - \phi_1 X_{t-1} - \cdots \phi_p X_{t-p} = Z_t + \theta_1 Z_{t-1} + \cdots + \theta_q Z_{t-q}$$

where $(Z_t) \sim \text{WN}(0, \sigma^2)$. We say that (X_t) is an ARMA(p,q) process with mean μ if $(X_t - \mu)$ is an ARMA(p,q) process.

The equation can be written as

$$\phi(B)X_t = \theta(B)Z_t$$

where ϕ, θ are p, q-degree polynomials and B is the push-backward operator.

Definiton 2.1.3 (The MA(q) process and the AR(p) process)**.** The MA(q) process is an ARMA$(0,q)$ process, that is $\phi \equiv 1$, i.e.

$$X_t = \theta(B)Z_t.$$

The AR(p) process is an ARMA$(p,0)$ process, that is $\theta \equiv 1$, i.e.

$$\phi(B)X_t = Z_t$$

Proposition 2.1.1. The mean value and covariance function for an MA(q) process X_t with MA polynomial θ is

$$\mu_X(t) = 0, \quad \gamma_X(h) = \sigma^2 \left(\sum_{k=0}^{q-h} \theta_k \theta_{k+h} \right).$$

Proof. We know

$$\mathbb{E}X_t = \mathbb{E}\theta(B)Z_t = 0$$

and

$$\gamma_X(h) = \mathbb{E}\left(\mathbb{E}\theta(Z)Z_t\right)\left(\theta(Z)Z_{t+h}\right) = \sigma^2 \left(\sum_{k=0}^{q-h} \theta_k \theta_{k+h} \right)$$

where $\theta_0 := 1$. □

Definiton 2.1.4 (Causality)**.** An ARMA(p,q) process defined by the equations $\phi(B)X_t = \theta(B)Z_t$ is said to be *causal* (or more specifically to be a *causal function of* $\{Z_t\}$) if there

exists a sequence of constants $\{\psi_j\}_{j\geq 0}$ such that

$$\sum_{j=0}^{\infty} |\psi_j| < \infty$$

and

$$X_t = \sum_{j=0}^{\infty} \psi_j Z_{t-j}, \qquad t = 0, \pm 1, \ldots.$$

Proposition 2.1.2. If $\{X_t\}_{t\in\mathbb{Z}}$ is any sequence of random variables such that $\sup_t \mathbb{E}|X_t| < \infty$, and if $\sum_{j=-\infty}^{\infty} |\psi_j| < \infty$, then the series

$$\psi(B)X_t = \sum_{j=-\infty}^{\infty} \psi_j B^j X_t = \sum_{j=-\infty}^{\infty} \psi_j X_{t-j}$$

converges absolutely with probability one. If in addition $\sup_t \mathbb{E}|X_t|^2 < \infty$, then the series converges in mean square to the same limit.

Proof. To see the absolute convergence for a.s., we may notice that

$$\mathbb{E}|\phi|(B)|X_t| \leq \sum_{j=-\infty}^{\infty} |\phi_j| \sup \mathbb{E}|X_t| < \infty$$

and hence $|\phi|(B)|X_t|$ has to be finite in probability 1.

Then we would like to show that the series converge to $\phi(B)X_t$ in L^2 as well, that is we may prove that by noticing

$$\mathbb{E}\left|\sum_{m<|j|\leq n} \phi_j X_{t-j}\right|^2 \leq \left(\sum_{|j|\geq m} |\phi_j|\right)^2 \sup_t \mathbb{E}|X_t|^2 \to 0$$

and hence the series converge in L^2 to some limit L, with

$$\mathbb{E}|L - \sum_{j=-\infty}^{\infty} \phi_j X_{t-j}|^2 \leq \liminf \mathbb{E}|L - \sum_{j=-n}^{n} \phi_j X_{t-j}|^2 = 0$$

by Fatou's lemma and we are done. □

Proposition 2.1.3. If $\{X_t\}$ is a stationary process with autocovariance function $\gamma(\cdot)$ and if $\sum_{j=-\infty}^{\infty} |\psi_j| < \infty$, then for each $t \in \mathbb{Z}$ the series

$$\psi(B)X_t = \sum_{j=-\infty}^{\infty} \psi_j X_{t-j}$$

converges absolutely with probability one and in mean square to the same limit. If

$$Y_t = \psi(B)X_t,$$

then the process $\{Y_t\}$ is stationary with autocovariance function

$$\gamma_Y(h) = \sum_{j,k=-\infty}^{\infty} \psi_j \psi_k \gamma(h - j + k).$$

Theorem 2.1.4. Let $\{X_t\}$ be an ARMA(p, q) process for which the polynomials $\phi(\cdot)$ and $\theta(\cdot)$ have no common zeroes. Then $\{X_t\}$ is causal if and only if $\phi(z) \neq 0$ for all $z \in \mathbb{C}$ such that $|z| \leq 1$. The coefficients $\{\psi_j\}$ in the causal representation $X_t = \sum_{j=0}^{\infty} \psi_j Z_{t-j}$ are determined by the relation

$$\psi(z) = \sum_{j=0}^{\infty} \psi_j z^j = \frac{\theta(z)}{\phi(z)}, \qquad |z| \leq 1.$$

Proof. Suppose first that $\phi(z) \neq 0$ for every $|z| \leq 1$. Since the zero set of the polynomial ϕ is finite, there is some $R > 1$ such that ϕ has no zero in $|z| < R$. Hence the rational function

$$\psi(z) := \frac{\theta(z)}{\phi(z)}$$

is analytic in $|z| < R$ and has a power-series expansion

$$\psi(z) = \sum_{j=0}^{\infty} \psi_j z^j.$$

Choose r with $1 < r < R$. Cauchy's estimate gives

$$|\psi_j| \leq M_r r^{-j}, \qquad M_r := \max_{|z|=r} |\psi(z)|,$$

so $\sum_{j=0}^{\infty} |\psi_j| < \infty$. Consequently

$$Y_t := \psi(B) Z_t = \sum_{j=0}^{\infty} \psi_j Z_{t-j}$$

is well defined, and the preceding propositions show that $\{Y_t\}$ is stationary. Since $\phi(z)\psi(z) = \theta(z)$, absolute summability permits composition of the filters and gives

$$\phi(B) Y_t = \theta(B) Z_t = \phi(B) X_t.$$

Put $U_t = X_t - Y_t$. Then $\{U_t\}$ is stationary and $\phi(B) U_t = 0$. The function $1/\phi(z)$ is also analytic in $|z| < R$ and has absolutely summable Taylor coefficients. Applying the corresponding filter to the last equality yields

$$U_t = \frac{1}{\phi}(B)\phi(B) U_t = 0.$$

Thus $X_t = Y_t = \sum_{j=0}^{\infty} \psi_j Z_{t-j}$, so $\{X_t\}$ is causal.

Conversely, suppose that $\{X_t\}$ is causal, so that

$$X_t = \psi(B)Z_t = \sum_{j=0}^{\infty} \psi_j Z_{t-j}, \qquad \sum_{j=0}^{\infty} |\psi_j| < \infty.$$

Substitution into the ARMA equation gives

$$\big(\phi(B)\psi(B) - \theta(B)\big)Z_t = 0.$$

Write

$$\phi(z)\psi(z) - \theta(z) = \sum_{j=0}^{\infty} a_j z^j.$$

The sequence $\{a_j\}$ is absolutely summable. Since $\{Z_t\}$ is white noise with variance $\sigma^2 > 0$,

$$0 = \mathrm{Var}\left(\sum_{j=0}^{\infty} a_j Z_{t-j}\right) = \sigma^2 \sum_{j=0}^{\infty} |a_j|^2.$$

Therefore $a_j = 0$ for every j, and hence

$$\phi(z)\psi(z) = \theta(z), \qquad |z| \le 1.$$

If $\phi(z_0) = 0$ for some $|z_0| \le 1$, the absolute convergence of $\psi(z_0)$ would imply $\theta(z_0) = 0$, contradicting the assumption that ϕ and θ have no common zeroes. Thus $\phi(z) \neq 0$ throughout $|z| \le 1$, and the displayed identity also gives $\psi(z) = \theta(z)/\phi(z)$. This relation uniquely determines all of the coefficients $\{\psi_j\}$. □

Definiton 2.1.5 (Invertibility)**.** An ARMA(p, q) process defined by the equations $\phi(B)X_t = \theta(B)Z_t$ is said to be *invertible* if there exists a sequence of constants $\{\pi_j\}_{j\ge 0}$ such that

$$\sum_{j=0}^{\infty} |\pi_j| < \infty$$

and

$$Z_t = \sum_{j=0}^{\infty} \pi_j X_{t-j}, \qquad t = 0, \pm 1, \ldots.$$

Theorem 2.1.5. Let $\{X_t\}$ be an ARMA(p, q) process for which the polynomials $\phi(\cdot)$ and $\theta(\cdot)$ have no common zeroes. Then $\{X_t\}$ is invertible if and only if $\theta(z) \neq 0$ for all $z \in \mathbb{C}$ such that $|z| \le 1$. The coefficients $\{\pi_j\}$ in the invertible representation $Z_t = \sum_{j=0}^{\infty} \pi_j X_{t-j}$ are determined by the relation

$$\pi(z) = \sum_{j=0}^{\infty} \pi_j z^j = \frac{\phi(z)}{\theta(z)}, \qquad |z| \le 1.$$

The coefficients $\{\pi_j\}$ can be calculated from recursion relations analogous to those for $\{\psi_j\}$.

Proof. □

3 Stationary ARMA Processes: Further Theory (Chapter 3)

This chapter continues the existing ARMA notes. The definitions and results on causality and invertibility already recorded there are not repeated. Throughout, $BX_t = X_{t-1}$,

$$\phi(z) = 1 - \sum_{j=1}^{p} \phi_j z^j, \qquad \theta(z) = 1 + \sum_{j=1}^{q} \theta_j z^j, \qquad \{Z_t\} \sim \mathrm{WN}(0, \sigma^2), \quad \sigma^2 > 0.$$

Unless stated otherwise, processes and coefficients are real, and processes are centered. All closed linear spans and projections below are in L^2.

3.1 Stationary Solutions and Laurent Expansions (Section 3.1)

Theorem 3.1.1 (Stationary solution; Theorem 3.1.3)**.** If $\phi(z) \neq 0$ for $|z| = 1$, the equations $\phi(B)X_t = \theta(B)Z_t$ have a unique stationary solution

$$X_t = \sum_{j \in \mathbb{Z}} \psi_j Z_{t-j}, \qquad \frac{\theta(z)}{\phi(z)} = \sum_{j \in \mathbb{Z}} \psi_j z^j \quad (r^{-1} < |z| < r)$$

for some $r > 1$. The coefficients are absolutely summable, and the random series converges absolutely almost surely and in mean square. Uniqueness is with respect to the given noise sequence, up to almost sure equality.

Proposition 3.1.2 (Unit-circle obstruction)**.** If ϕ and θ have no common zeroes and ϕ has a zero on the unit circle, there is no stationary solution with finite second moments. Common factors must therefore be removed before applying this criterion.

Corollary 3.1.3 (Geometric decay)**.** For the solution above, there exist $K < \infty$ and $a \in (0, 1)$ such that

$$|\psi_j| \leq K a^{|j|}, \qquad j \in \mathbb{Z}.$$

There also exist $C < \infty$ and $s \in (0, 1)$ such that

$$|\gamma(h)| \leq C s^{|h|}, \qquad \sum_{h \in \mathbb{Z}} |\gamma(h)| < \infty.$$

In general, a stationary representation is bilateral; causality requires the negative-index coefficients to vanish.

3.2 Infinite Moving Averages (Section 3.2)

Definiton 3.2.1 (Moving average of infinite order)**.** An MA(∞) of $\{Z_t\}$ is a process of the form

$$X_t = \sum_{j=0}^{\infty} \psi_j Z_{t-j}, \qquad \sum_{j=0}^{\infty} |\psi_j| < \infty.$$

The one-sided index set is part of this definition.

Theorem 3.2.1 (Covariance of an infinite moving average)**.** The process above is stationary,

with mean zero and

$$\gamma(h) = \sigma^2 \sum_{j=0}^{\infty} \psi_j \psi_{j+|h|}.$$

More generally, $\sum_{j\geq 0} \psi_j^2 < \infty$ suffices for mean-square convergence and the same covariance formula, although it does not imply absolute summability of the coefficients.

Proposition 3.2.2 (Finite covariance range; Proposition 3.2.1)**.** If a zero-mean stationary process satisfies $\gamma(h) = 0$ for $|h| > q$ and $\gamma(q) \neq 0$, then there is a white noise sequence $\{\varepsilon_t\}$ such that

$$X_t = \varepsilon_t + \vartheta_1 \varepsilon_{t-1} + \cdots + \vartheta_q \varepsilon_{t-q}.$$

It can be chosen as the innovation $\varepsilon_t = X_t - P_{\mathcal{M}_{t-1}} X_t$, where $\mathcal{M}_t = \overline{\operatorname{span}}\{X_s : s \leq t\}$.

Corollary 3.2.3 (Second-order characterization)**.** If a zero-mean stationary process has the same autocovariance function as an ARMA(p, q) process, it too admits an ARMA(p, q) representation with some white noise sequence. Equality of covariance functions does not imply equality of finite-dimensional distributions.

3.3 Coefficient and Covariance Identities (Section 3.3)

For a causal ARMA process, put $\psi_j = 0$ for $j < 0$, $\theta_0 = 1$, and $\theta_j = 0$ for $j > q$. The identity $\phi(z)\psi(z) = \theta(z)$ gives

$$\psi_0 = 1, \qquad \psi_j = \theta_j + \sum_{i=1}^{\min(p,j)} \phi_i \psi_{j-i}, \quad j \geq 1.$$

Its covariance function satisfies

$$\gamma(k) - \sum_{i=1}^{p} \phi_i \gamma(k-i) = \sigma^2 \sum_{j=k}^{q} \theta_j \psi_{j-k}, \qquad k \geq 0,$$

where an empty sum is zero and $\gamma(-k) = \gamma(k)$. In particular,

$$\gamma(k) = \sum_{i=1}^{p} \phi_i \gamma(k-i), \quad k > q.$$

For an AR(p) process these specialize to the Yule–Walker identities

$$\gamma(0) - \sum_{i=1}^{p} \phi_i \gamma(i) = \sigma^2, \qquad \gamma(k) = \sum_{i=1}^{p} \phi_i \gamma(k-i), \quad k \geq 1.$$

These are population identities; no parameter estimation is being assumed.

3.4 Partial Autocorrelation (Section 3.4)

Definiton 3.4.1 (Partial autocorrelation function)**.** Let $\gamma(0) > 0$ and $\rho(h) = \gamma(h)/\gamma(0)$. Set $\alpha(1) = \rho(1)$. For $k \geq 2$, let $\mathcal{H}_k = \overline{\operatorname{span}}\{X_2, \ldots, X_k\}$ and define

$$\alpha(k) = \operatorname{Corr}\big(X_{k+1} - P_{\mathcal{H}_k} X_{k+1},\, X_1 - P_{\mathcal{H}_k} X_1\big),$$

provided the residual variances are positive. For nonzero means, include the constant 1 in the projecting space, or first center the process.

Proposition 3.4.1 (Prediction-coefficient characterization)**.** Suppose the finite covariance matrices are nonsingular. If

$$P_{\mathrm{span}\{X_1,\dots,X_k\}}X_{k+1} = \sum_{j=1}^{k} \phi_{kj} X_{k+1-j},$$

then

$$\alpha(k) = \phi_{kk}, \qquad R_k \boldsymbol{\phi}_k = \boldsymbol{\rho}_k, \quad R_k = [\rho(i-j)]_{i,j=1}^{k}, \quad \boldsymbol{\rho}_k = (\rho(1),\dots,\rho(k))^\top.$$

The sample PACF is defined by replacing ρ by the sample ACF in these equations when the resulting system has a unique solution.

Proposition 3.4.2 (Cutoff properties)**.** For a causal AR(p) process, $\alpha(k) = 0$ for $k > p$; if its order is exactly p, then $\alpha(p) = \phi_p \neq 0$. For an MA($q$) process, $\gamma(h) = 0$ for $|h| > q$. Thus AR order appears as a PACF cutoff, and MA order as an ACF cutoff.

3.5 Autocovariance Generating Functions (Section 3.5)

Definiton 3.5.1 (Autocovariance generating function)**.** When the Laurent series converges on an annulus containing the unit circle, define

$$G(z) = \sum_{h\in\mathbb{Z}} \gamma(h) z^h.$$

For real-valued processes $G(z) = G(z^{-1})$ wherever both sides are defined.

Proposition 3.5.1 (Filtering identity)**.** If $X_t = \sum_{j\in\mathbb{Z}} \psi_j Z_{t-j}$ and $\psi(z) = \sum_j \psi_j z^j$ converges absolutely on a symmetric annulus $r^{-1} < |z| < r$, then

$$G(z) = \sigma^2 \psi(z)\psi(z^{-1}).$$

Consequently, for stationary ARMA equations with no AR zero on the unit circle,

$$G(z) = \sigma^2 \frac{\theta(z)\theta(z^{-1})}{\phi(z)\phi(z^{-1})}$$

on a sufficiently small such annulus. A process is white noise precisely when its covariance generating function is constant.

Proposition 3.5.2 (Causal invertible representation; Proposition 3.5.1)**.** If neither ϕ nor θ has a zero on the unit circle, the same process admits a representation

$$\widetilde{\phi}(B) X_t = \widetilde{\theta}(B) \widetilde{Z}_t,$$

where $\widetilde{\phi}$ and $\widetilde{\theta}$ have no zeroes in $|z| \leq 1$. The degrees can be retained. This is obtained by reflecting roots inside the unit circle to their reciprocal conjugates and adjusting the noise variance. The new noise is white; independence need not be preserved unless Gaussianity is assumed. Causality and invertibility concern the specified noise representation.

3.6 Linear Difference Equations (Section 3.6)

Theorem 3.6.1 (General homogeneous solution)**.** Let $a(z) = 1 + a_1 z + \cdots + a_k z^k$, $a_k \neq 0$, have distinct zeroes $\zeta_1, \ldots, \zeta_r$ of multiplicities $m_1, \ldots, m_r$. The solution space of $a(B)u_t = 0$ has dimension k, with basis

$$\{t^\ell \zeta_i^{-t} : 1 \leq i \leq r,\ 0 \leq \ell < m_i\}.$$

Thus

$$u_t = \sum_{i=1}^{r} \sum_{\ell=0}^{m_i - 1} c_{i\ell} t^\ell \zeta_i^{-t}.$$

Any k consecutive initial values determine a unique solution. For real solutions the coefficients attached to conjugate roots are conjugates.

Complex conjugate roots give polynomially modulated sinusoids. If every $|\zeta_i| > 1$, then $|u_t| \leq C s^t$ for $t \geq 0$ and some $s \in (0, 1)$. The same root structure describes the eventual form of causal ARMA coefficients and autocovariances.

4 The Spectral Representation of a Stationary Process (Chapter 4)

The frequency domain represents covariance by a finite positive measure and represents the process itself by a mean-square stochastic integral. The convention throughout is

$$\gamma(h) = \int_{(-\pi,\pi]} e^{ih\lambda}\, dF(\lambda).$$

Thus white noise of variance σ^2 has density $\sigma^2/(2\pi)$. All stochastic integrals in this chapter have deterministic integrands.

4.1 Complex Stationarity and Spectral Measures (Sections 4.1–4.3)

Definiton 4.1.1 (Complex-valued weak stationarity)**.** A complex process is weakly stationary if $\mathbb{E}|X_t|^2 < \infty$, $\mathbb{E}X_t = \mu$ is constant, and

$$\gamma(h) = \mathbb{E}[(X_{t+h} - \mu)\overline{(X_t - \mu)}]$$

depends only on h. We use $\langle U, V\rangle = \mathbb{E}[U\overline{V}]$. Then $\gamma(-h) = \overline{\gamma(h)}$ and the covariance matrices are Hermitian and nonnegative definite.

Theorem 4.1.1 (Herglotz; Theorem 4.3.1)**.** A function $\gamma : \mathbb{Z} \to \mathbb{C}$ is an autocovariance function if and only if it is nonnegative definite, equivalently if and only if there is a finite positive measure F on $(-\pi, \pi]$ such that

$$\gamma(h) = \int_{(-\pi,\pi]} e^{ih\lambda}\, dF(\lambda), \qquad h \in \mathbb{Z}.$$

Nonnegative definiteness means $\sum_{i,j=1}^{n} a_i \overline{a_j} \gamma(i-j) \geq 0$ for all n and $a_1, \ldots, a_n \in \mathbb{C}$. The spectral measure is unique. Its distribution function is normalized by $F(-\pi) = 0$ and is bounded, nondecreasing and right-continuous, with $F(\pi) = \gamma(0)$.

Definiton 4.1.2 (Spectral density and spectral atoms)**.** If $dF(\lambda) = f(\lambda)\, d\lambda$, then f is a spectral density. It is nonnegative and integrable, and is defined up to Lebesgue-null sets. An atom of mass a at λ_0 contributes $ae^{ih\lambda_0}$ to $\gamma(h)$. Spectral distributions need not have densities. For real-valued processes, the spectral measure is symmetric on the circle; when a density exists it is even almost everywhere, and

$$\gamma(h) = 2\int_0^{\pi} f(\lambda)\cos(h\lambda)\, d\lambda.$$

Theorem 4.1.2 (Absolutely summable covariances; Theorem 4.3.2)**.** If $\sum_{h\in\mathbb{Z}} |\gamma(h)| < \infty$, then

$$f(\lambda) = \frac{1}{2\pi} \sum_{h\in\mathbb{Z}} \gamma(h) e^{-ih\lambda}$$

converges absolutely and uniformly to a continuous spectral density. Conversely, an absolutely summable sequence is an autocovariance function if and only if its Fourier series above is nonnegative at every frequency.

4.2 Filtering, ARMA Spectra and Approximation (Section 4.4)

Theorem 4.2.1 (Spectral effect of an absolutely summable filter)**.** Let $X_t = \sum_{j\in\mathbb{Z}} \psi_j Y_{t-j}$, where Y is centered and stationary and $\sum_j |\psi_j| < \infty$. Define $\Psi(\lambda) = \sum_j \psi_j e^{-ij\lambda}$. Then X is stationary and

$$dF_X(\lambda) = |\Psi(\lambda)|^2 dF_Y(\lambda), \qquad f_X(\lambda) = |\Psi(\lambda)|^2 f_Y(\lambda)$$

when Y has a density. Ψ is the transfer function and $|\Psi|^2$ is the power transfer function. For complex coefficients,

$$\gamma_X(h) = \sum_{j,k\in\mathbb{Z}} \psi_j \overline{\psi_k} \gamma_Y(h - j + k).$$

Theorem 4.2.2 (ARMA spectral density; Theorem 4.4.2)**.** For $\phi(B)X_t = \theta(B)Z_t$, with $\{Z_t\} \sim \mathrm{WN}(0, \sigma^2)$ and $\phi(z) \neq 0$ on $|z| = 1$,

$$f_X(\lambda) = \frac{\sigma^2}{2\pi} \frac{|\theta(e^{-i\lambda})|^2}{|\phi(e^{-i\lambda})|^2}, \qquad -\pi \leq \lambda \leq \pi.$$

This holds for the stationary solution even if its representation is noncausal or noninvertible. The density is rational in $e^{i\lambda}$.

Proposition 4.2.3 (Recovery of the noise in a closed span)**.** Assume ϕ, θ are coprime and ϕ has no zero on the unit circle. Then

$$Z_t \in \overline{\mathrm{span}}\{X_s : s \leq t\} \quad \Longleftrightarrow \quad \theta(z) \neq 0 \text{ for } |z| < 1.$$

This closed-span notion permits zeroes on $|z| = 1$; the absolutely summable inverse used earlier requires no zeroes on $|z| \leq 1$. If also ϕ has no zeroes on $|z| \leq 1$, the recovered noise is the innovation $Z_t = X_t - P_{\mathcal{M}_{t-1}} X_t$.

Theorem 4.2.4 (Uniform spectral approximation; Section 4.4)**.** For every real symmetric continuous spectral density f and every $\epsilon > 0$, there exist an invertible finite MA process and a causal finite AR process, each with density g satisfying

$$\sup_{\lambda\in[-\pi,\pi]} |f(\lambda) - g(\lambda)| < \epsilon.$$

The orders are allowed to depend on f and ϵ.

4.3 Covariance Matrices and Fourier Coordinates (Section 4.5)

Proposition 4.3.1 (Circulant eigenstructure)**.** Let $C_{rs} = c_{(s-r) \bmod n}$ for $0 \leq r, s < n$ and $\omega_k = 2\pi k/n$. The vectors

$$v_k = n^{-1/2}(1, e^{i\omega_k}, \ldots, e^{i(n-1)\omega_k})^\top$$

form an orthonormal eigenbasis, with eigenvalues $\lambda_k = \sum_{j=0}^{n-1} c_j e^{ij\omega_k}$. If C is Hermitian, all λ_k are real.

Proposition 4.3.2 (Spectral quadratic form and eigenvalue bounds)**.** For $\Gamma_n = [\gamma(i -$

$j)]_{i,j=1}^n$ and $a \in \mathbb{C}^n$,

$$a^* \Gamma_n a = \int_{-\pi}^{\pi} \left| \sum_{j=1}^{n} a_j e^{-ij\lambda} \right|^2 f(\lambda)\, d\lambda.$$

If $0 < m \le f \le M < \infty$ almost everywhere, then

$$2\pi m \|a\|^2 \le a^* \Gamma_n a \le 2\pi M \|a\|^2.$$

Every eigenvalue of Γ_n lies in $[2\pi m, 2\pi M]$.

Proposition 4.3.3 (Second moments of Fourier coordinates)**.** For a real centered process with absolutely summable covariances, define

$$D_n(\lambda) = n^{-1/2} \sum_{t=0}^{n-1} X_t e^{-it\lambda}.$$

Then

$$\mathbb{E}|D_n(\lambda)|^2 = \sum_{|h|<n} \left(1 - \frac{|h|}{n}\right) \gamma(h) e^{-ih\lambda} \longrightarrow 2\pi f(\lambda)$$

uniformly in λ. At the Fourier frequencies $2\pi k/n$, distinct coordinates are asymptotically uncorrelated in the Hermitian covariance sense. Uncorrelatedness alone does not imply independence.

4.4 Orthogonal Increments and Stochastic Integration (Sections 4.6–4.7)

Definiton 4.4.1 (Orthogonal-increment process)**.** A centered complex process $Z(\lambda)$, $-\pi \le \lambda \le \pi$, has orthogonal increments if its increments over disjoint intervals are orthogonal in L^2. Take $Z(-\pi) = 0$ and assume right-continuity in mean square. Its associated finite measure is specified by

$$\mathbb{E}|Z(b) - Z(a)|^2 = F(b) - F(a), \quad -\pi \le a < b \le \pi.$$

More generally the covariance of two increments is the F-measure of the intersection of their intervals. Orthogonal increments need not be independent.

Definiton 4.4.2 (Integral with respect to orthogonal increments)**.** For an interval step function $g = \sum_j c_j \mathbf{1}_{(a_j, b_j]}$, define

$$I(g) = \int g\, dZ = \sum_j c_j [Z(b_j) - Z(a_j)].$$

For arbitrary $g \in L^2(F)$, choose step functions $g_n \to g$ in $L^2(F)$ and define $I(g) = \lim_n I(g_n)$ in mean square. This definition is independent of the approximating sequence.

Theorem 4.4.1 (Isometry)**.** For $g, h \in L^2(F)$ and $a, b \in \mathbb{C}$,

$$I(ag + bh) = aI(g) + bI(h), \qquad \mathbb{E}I(g) = 0,$$

$$\mathbb{E}[I(g)\overline{I(h)}] = \int g\bar{h}\, dF, \qquad \mathbb{E}|I(g)|^2 = \int |g|^2\, dF.$$

Thus $g_n \to g$ in $L^2(F)$ if and only if $I(g_n) \to I(g)$ in mean square.

4.5 Spectral Representation and Inversion (Sections 4.8–4.9)

Theorem 4.5.1 (Spectral representation; Theorem 4.8.2)**.** Every centered stationary process admits the representation

$$X_t = \int_{(-\pi,\pi]} e^{it\lambda}\, dZ(\lambda), \qquad t \in \mathbb{Z},$$

where Z has orthogonal increments and associated measure F, the spectral measure of X. With $Z(-\pi) = 0$, $Z(\lambda)$ is unique up to almost sure equality for each fixed λ. The map I is a unitary correspondence

$$L^2(F) \longrightarrow \mathcal{H}_X := \overline{\text{span}}\{X_t : t \in \mathbb{Z}\}, \qquad e^{it\lambda} \longmapsto X_t.$$

Proposition 4.5.2 (A spectral atom as a random sinusoid)**.** If F has an atom at λ_0, put $\Delta Z(\lambda_0) = Z(\lambda_0) - Z(\lambda_0-)$. Then

$$X_t = e^{it\lambda_0}\Delta Z(\lambda_0) + \int_{(-\pi,\pi]\setminus\{\lambda_0\}} e^{it\lambda}\, dZ(\lambda),$$

where the two components are orthogonal and $\mathbb{E}|\Delta Z(\lambda_0)|^2 = \Delta F(\lambda_0)$. For real processes, conjugate frequency pairs combine into real sinusoidal components.

Theorem 4.5.3 (Inversion at continuity points; Theorem 4.9.1)**.** If $-\pi < v < w < \pi$ are continuity points of F, then

$$\frac{1}{2\pi}\sum_{|j|\le n} X_j \int_v^w e^{-ij\lambda}\, d\lambda \xrightarrow{L^2} Z(w) - Z(v),$$

$$\frac{1}{2\pi}\sum_{|j|\le n} \gamma(j) \int_v^w e^{-ij\lambda}\, d\lambda \longrightarrow F(w) - F(v).$$

The endpoint continuity condition excludes spectral atoms at v and w.

Theorem 4.5.4 (Continuous-time analogue; Theorem 4.9.2)**.** A centered stationary process indexed by $\mathbb{R}$, whose covariance is continuous at zero, has a finite spectral measure on $\mathbb{R}$ with

$$\gamma(s) = \int_{\mathbb{R}} e^{is\lambda}\, dF(\lambda), \qquad X_t = \int_{\mathbb{R}} e^{it\lambda}\, dZ(\lambda).$$

At continuity points $v < w$, the analogous inversion formulae are

$$\frac{1}{2\pi}\int_{-T}^{T} X_t \left(\int_v^w e^{-it\lambda}\, d\lambda\right) dt \xrightarrow{L^2} Z(w) - Z(v),$$

with X_t replaced by $\gamma(t)$ for recovery of $F(w) - F(v)$.

4.6 General Linear Filters and Fourier Approximation (Sections 4.10–4.11)

Theorem 4.6.1 (Mean-square filtering; Theorem 4.10.1)**.** Let $H_n(\lambda) = \sum_{j=-n}^{n} h_j e^{-ij\lambda}$ converge to H in $L^2(F_X)$. Then

$$Y_t = \lim_{n\to\infty} \sum_{j=-n}^{n} h_j X_{t-j} = \int e^{it\lambda} H(\lambda)\, dZ_X(\lambda)$$

exists in mean square, is stationary, and has spectral measure $dF_Y = |H|^2 dF_X$. If $H \neq 0$ F_X-almost everywhere, then

$$X_t = \int e^{it\lambda} H(\lambda)^{-1}\, dZ_Y(\lambda), \qquad dZ_Y = H\, dZ_X.$$

In particular, X_t belongs to the closed span of the entire output process. A filter is causal when $h_j = 0$ for $j < 0$.

Proposition 4.6.2 (Fourier approximation of an interval indicator)**.** For $-\pi < v < w < \pi$, set

$$a_j = \frac{1}{2\pi} \int_v^w e^{-ij\lambda}\, d\lambda, \qquad h_n(\lambda) = \sum_{|j|\le n} a_j e^{ij\lambda}.$$

The functions h_n are uniformly bounded in n and λ and converge uniformly to $\mathbf{1}_{(v,w]}$ outside any open neighborhoods of the two endpoints. If F has no atoms at these endpoints, then $h_n \to \mathbf{1}_{(v,w]}$ in $L^2(F)$. This is the approximation underlying the inversion formulae above.

5 Prediction of Stationary Processes (Chapter 5)

Unless otherwise stated, X is a real, centered stationary process with covariance γ. Write

$$\mathcal{H}_n = \operatorname{span}\{X_1, \dots, X_n\}, \qquad \mathcal{M}_n = \overline{\operatorname{span}}\{X_t : t \le n\}.$$

$P_{\mathcal{H}}$ always denotes orthogonal projection in L^2. For a nonzero known mean μ, apply the results to $X_t - \mu$ and add μ to the predicted value.

5.1 Prediction Equations and Optimality (Section 5.1)

Definiton 5.1.1 (Best linear predictor)**.** The best linear predictor of X_{n+h} from $X_1, \dots, X_n$ is

$$\widehat{X}_{n+h|n} = P_{\mathcal{H}_n} X_{n+h}.$$

It uniquely minimizes $\mathbb{E}(X_{n+h} - Y)^2$ over $Y \in \mathcal{H}_n$ and is characterized by

$$\mathbb{E}[(X_{n+h} - \widehat{X}_{n+h|n})Y] = 0, \qquad Y \in \mathcal{H}_n.$$

Theorem 5.1.1 (Normal equations)**.** Let $\Gamma_n = [\gamma(i-j)]_{i,j=1}^n$ and $\boldsymbol{\gamma}_n^{(h)} = (\gamma(h), \dots, \gamma(h+n-1))^\top$. Then

$$\widehat{X}_{n+h|n} = \sum_{j=1}^{n} \phi_{nj}^{(h)} X_{n+1-j}, \qquad \Gamma_n \boldsymbol{\phi}_n^{(h)} = \boldsymbol{\gamma}_n^{(h)}.$$

A solution always exists. The predictor is unique even when its coefficient vector is not. If Γ_n is nonsingular, then

$$\boldsymbol{\phi}_n^{(h)} = \Gamma_n^{-1} \boldsymbol{\gamma}_n^{(h)}, \qquad v_n(h) = \gamma(0) - (\boldsymbol{\gamma}_n^{(h)})^\top \Gamma_n^{-1} \boldsymbol{\gamma}_n^{(h)}.$$

Here $v_n(h) = \mathbb{E}(X_{n+h} - \widehat{X}_{n+h|n})^2$.

Proposition 5.1.2 (Nonsingularity; Proposition 5.1.1)**.** If $\gamma(0) > 0$ and $\gamma(h) \to 0$ as $|h| \to \infty$, then Γ_n is nonsingular for every n. This is sufficient, not necessary.

For every $Y \in \mathcal{H}_n$, the orthogonal decomposition gives

$$\mathbb{E}(X_{n+h} - Y)^2 = v_n(h) + \mathbb{E}(\widehat{X}_{n+h|n} - Y)^2.$$

Thus the normal equations express an optimization property, rather than merely a procedure for finding coefficients.

5.2 Recursive Prediction and Innovations (Section 5.2)

Theorem 5.2.1 (Durbin–Levinson recursion; Proposition 5.2.1)**.** Suppose all Γ_n are positive definite. For one-step prediction write $\phi_{nj} = \phi_{nj}^{(1)}$ and $v_n = v_n(1)$, with $v_0 = \gamma(0)$. Then $\phi_{11} = \gamma(1)/\gamma(0)$ and, for $n \ge 2$,

$$\kappa_n := \phi_{nn} = \frac{\gamma(n) - \sum_{j=1}^{n-1} \phi_{n-1,j}\gamma(n-j)}{v_{n-1}},$$

$$\phi_{nj} = \phi_{n-1,j} - \kappa_n \phi_{n-1,n-j}, \qquad 1 \le j < n,$$

$$v_n = v_{n-1}(1-\kappa_n^2), \qquad n \ge 1, \quad \kappa_1 = \phi_{11}.$$

The reflection coefficient κ_n is the partial autocorrelation $\alpha(n)$. Positive definiteness implies $|\kappa_n| < 1$.

Corollary 5.2.2 (Prediction variance and determinant)**.**

$$v_n = \gamma(0)\prod_{j=1}^{n}(1-\alpha(j)^2), \qquad \det \Gamma_n = \prod_{j=0}^{n-1} v_j.$$

The decreasing sequence v_n is the error remaining after increasingly long observation histories.

Definiton 5.2.1 (Finite-history innovations)**.** Set $\widehat{X}_{1|0} = 0$ and

$$e_t = X_t - \widehat{X}_{t|t-1}, \qquad \mathbb{E}e_t^2 = v_{t-1}.$$

The e_t are pairwise orthogonal and $\mathcal{H}_n = \operatorname{span}\{e_1, \dots, e_n\}$. They need not be independent or have equal variances.

Theorem 5.2.3 (Innovations representation and recursion)**.** Assume $v_j > 0$. The one-step predictor can be written

$$\widehat{X}_{n+1|n} = \sum_{j=1}^{n} \vartheta_{nj} e_{n+1-j}.$$

With $v_0 = \gamma(0)$, the coefficients satisfy, for $0 \le k < n$,

$$\vartheta_{n,n-k} = \frac{1}{v_k}\left(\gamma(n-k) - \sum_{j=0}^{k-1} \vartheta_{k,k-j}\vartheta_{n,n-j}v_j\right),$$

$$v_n = \gamma(0) - \sum_{j=0}^{n-1} \vartheta_{n,n-j}^2 v_j.$$

An empty sum is zero. Unlike Durbin–Levinson, which updates coefficients of observations, this recursion updates coefficients of orthogonal innovations.

The innovations recursion also applies to a centered nonstationary sequence. Writing $K(s,t) = \mathbb{E}(X_s X_t)$ and assuming every finite covariance matrix is nonsingular, its general form is

$$\vartheta_{n,n-k} = v_k^{-1}\left(K(n+1,k+1) - \sum_{j=0}^{k-1} \vartheta_{k,k-j}\vartheta_{n,n-j}v_j\right),$$

$$v_0 = K(1,1), \qquad v_n = K(n+1,n+1) - \sum_{j=0}^{n-1} \vartheta_{n,n-j}^2 v_j.$$

Stationarity reduces $K(n+1,k+1)$ to $\gamma(n-k)$.

Proposition 5.2.4 (Multiple-step prediction in innovations coordinates)**.** For $h \ge 1$,

$$\widehat{X}_{n+h|n} = \sum_{j=1}^{n} \frac{\mathbb{E}(X_{n+h}e_j)}{v_{j-1}} e_j, \qquad v_n(h) = \gamma(0) - \sum_{j=1}^{n} \frac{[\mathbb{E}(X_{n+h}e_j)]^2}{v_{j-1}}.$$

Writing $\mathbf{X}_n = (X_1, \dots, X_n)^\top$ and $X_t = \sum_{j=1}^{t} L_{tj} e_j$, with $L_{tt} = 1$, gives the factorization

$$\Gamma_n = L \operatorname{diag}(v_0, \dots, v_{n-1}) L^\top.$$

5.3 Finite-History ARMA Prediction (Section 5.3)

Let X be causal ARMA(p, q), and put $\widehat{X}_{s|n} = X_s$ for $1 \le s \le n$.

Proposition 5.3.1 (ARMA prediction equations)**.** For $n + h > \max(p, q)$, projection of the ARMA equation gives

$$\widehat{X}_{n+h|n} = \sum_{i=1}^{p} \phi_i \widehat{X}_{n+h-i|n} + \sum_{j=h}^{q} \theta_j P_{\mathcal{H}_n} Z_{n+h-j}.$$

In particular, if $h > q$, the noise term vanishes and the forecasts satisfy the homogeneous AR difference equation. Finite-history projections of Z_s need not equal Z_s, even when the model is invertible.

Proposition 5.3.2 (ARMA innovations form; Section 5.3)**.** Put $m = \max(p, q)$ and introduce the triangularly transformed observations

$$Y_t = \begin{cases} X_t, & 1 \le t \le m, \\ \phi(B) X_t, & t > m. \end{cases}$$

For every n, $\operatorname{span}\{Y_1, \dots, Y_n\} = \mathcal{H}_n$, and the finite-history innovations of Y are precisely e_t. Apply the general innovations recursion to the covariance of Y, writing its coefficients as η_{nj}. For $n \ge m$,

$$\widehat{X}_{n+1|n} = \sum_{i=1}^{p} \phi_i X_{n+1-i} + \sum_{j=1}^{q} \eta_{nj} e_{n+1-j}.$$

For $n + h > m$, the multiple-step version is

$$\widehat{X}_{n+h|n} = \sum_{i=1}^{p} \phi_i \widehat{X}_{n+h-i|n} + \sum_{j=h}^{\min(q, n+h-1)} \eta_{n+h-1, j} e_{n+h-j}.$$

Here $\widehat{X}_{s|n} = X_s$ for observed indices. The coefficients η_{nj} refer to Y, and generally differ from the coefficients ϑ_{nj} obtained by applying the recursion directly to X.

For arbitrary horizons, the normal equations give the exact finite-history error covariance. With $\mathbf{c}_h = (\gamma(n + h - 1), \dots, \gamma(h))^\top$, it is

$$\operatorname{Cov}(X_{n+h} - \widehat{X}_{n+h|n}, X_{n+k} - \widehat{X}_{n+k|n}) = \gamma(h - k) - \mathbf{c}_h^\top \Gamma_n^{-1} \mathbf{c}_k.$$

This formula applies to all stationary processes with nonsingular Γ_n, including ARMA models.

5.4 Gaussian Prediction and Prediction Bounds (Section 5.4)

Theorem 5.4.1 (Conditional Gaussian law)**.** For a centered stationary Gaussian process with nonsingular Γ_n,

$$\mathbb{E}[X_{n+h} \mid X_1, \dots, X_n] = \widehat{X}_{n+h|n}.$$

The prediction error is independent of $(X_1, \dots, X_n)$ and normally distributed with variance $v_n(h)$. If $v_n(h) > 0$, then conditionally on the observed history,

$$X_{n+h} \sim N(\widehat{X}_{n+h|n}, v_n(h)).$$

Hence, for $0 < \beta < 1$, an exact conditional $(1-\beta)$ prediction interval is

$$\widehat{X}_{n+h|n} \;\pm\; z_{1-\beta/2}\sqrt{v_n(h)}.$$

These statements assume the model parameters are known.

For a general non-Gaussian process, the best linear predictor need not equal the conditional expectation, which minimizes mean squared error over all square-integrable functions of the observations.

5.5 Prediction from the Infinite Past (Section 5.5)

Theorem 5.5.1 (Causal invertible ARMA prediction; Theorem 5.5.1)**.** Suppose

$$X_t = \sum_{j=0}^{\infty} \psi_j Z_{t-j}, \qquad Z_t = \sum_{j=0}^{\infty} \pi_j X_{t-j}, \qquad \psi_0 = \pi_0 = 1,$$

with both coefficient sequences absolutely summable. Then $\mathcal{M}_n = \overline{\operatorname{span}}\{Z_t : t \le n\}$, and

$$\widetilde{X}_{n+h|n} := P_{\mathcal{M}_n} X_{n+h} = \sum_{j=h}^{\infty} \psi_j Z_{n+h-j}.$$

The error and its variance are

$$E_{n,h} = \sum_{j=0}^{h-1} \psi_j Z_{n+h-j}, \qquad v_\infty(h) = \sigma^2 \sum_{j=0}^{h-1} \psi_j^2.$$

Writing $\widetilde{X}_{s|n} = X_s$ for $s \le n$ also gives

$$\widetilde{X}_{n+h|n} = -\sum_{j=1}^{\infty} \pi_j \widetilde{X}_{n+h-j|n}.$$

Proposition 5.5.2 (Joint forecast errors)**.** For $h, k \ge 1$,

$$\operatorname{Cov}(E_{n,h}, E_{n,k}) = \sigma^2 \sum_{\ell=1}^{\min(h,k)} \psi_{h-\ell}\psi_{k-\ell}.$$

Thus forecast errors at different horizons generally are correlated. Moreover $v_\infty(h) \uparrow \gamma(0)$ as $h \to \infty$.

Increasing finite histories ending at a fixed time yield projections that converge in L^2 to $P_{\mathcal{M}_n} X_{n+h}$. By stationarity, $v_n(h) \downarrow v_\infty(h)$ as the number of observed past values tends to infinity, for each fixed h.

5.6 Prediction in the Frequency Domain (Section 5.6)

Theorem 5.6.1 (Projection under the spectral isometry)**.** Let $I(g) = \int g\, dZ$ be the spectral isometry and set

$$\mathcal{K}_n = \overline{\text{span}}^{L^2(F)}\{e^{it\lambda} : t \le n\}.$$

Then

$$P_{\mathcal{M}_n} X_{n+h} = I(P_{\mathcal{K}_n} e^{i(n+h)\lambda}),$$

$$\mathbb{E}|X_{n+h} - P_{\mathcal{M}_n} X_{n+h}|^2 = \inf_{g \in \mathcal{K}_n} \int |e^{i(n+h)\lambda} - g(\lambda)|^2\, dF(\lambda).$$

Time-domain prediction is therefore equivalent to weighted L^2 approximation of exponentials by past-frequency functions.

For a causal invertible ARMA process, another expression for the predictor is

$$P_{\mathcal{M}_n} X_{n+h} = \sum_{j=0}^{\infty} a_j X_{n-j}, \qquad \sum_{j=0}^{\infty} a_j z^j = \frac{\phi(z)}{\theta(z)} \sum_{j=0}^{\infty} \psi_{j+h} z^j.$$

5.7 The Wold Decomposition (Section 5.7)

Definiton 5.7.1 (Determinism and pure nondeterminism)**.** Let $\mathcal{M} = \overline{\text{span}}\{X_t : t \in \mathbb{Z}\}$ and $\mathcal{M}_{-\infty} = \bigcap_{n \in \mathbb{Z}} \mathcal{M}_n$. The process is deterministic if

$$\sigma_\varepsilon^2 := \mathbb{E}(X_t - P_{\mathcal{M}_{t-1}} X_t)^2 = 0,$$

equivalently $X_t \in \mathcal{M}_{-\infty}$ for every t. It is purely nondeterministic if $\mathcal{M}_{-\infty} = \{0\}$. These are statements about linear predictability, not about sample paths being nonrandom.

Theorem 5.7.1 (Wold; Theorem 5.7.1)**.** If $\sigma_\varepsilon^2 > 0$, there is a unique decomposition

$$X_t = U_t + V_t, \qquad U_t = \sum_{j=0}^{\infty} \psi_j \varepsilon_{t-j},$$

where the series converges in mean square and

$$\psi_0 = 1, \quad \sum_{j=0}^{\infty} \psi_j^2 < \infty, \qquad \varepsilon_t = X_t - P_{\mathcal{M}_{t-1}} X_t \sim \text{WN}(0, \sigma_\varepsilon^2).$$

The coefficients and deterministic component are

$$\psi_j = \sigma_\varepsilon^{-2} \mathbb{E}(X_t \varepsilon_{t-j}), \qquad V_t = P_{\mathcal{M}_{-\infty}} X_t.$$

$\{V_t\}$ is stationary and deterministic, and $\mathbb{E}(V_s \varepsilon_t) = 0$ for all s, t. Furthermore,

$$\mathcal{M}_t = \overline{\text{span}}\{\varepsilon_s : s \le t\} \oplus \mathcal{M}_{-\infty}, \qquad \overline{\text{span}}\{V_s : s \le t\} = \mathcal{M}_{-\infty}.$$

Wold coefficients need only be square summable. The representation is therefore more general than the absolutely summable MA(∞) definition in Chapter 3.

Corollary 5.7.2 (Prediction from the Wold representation)**.**

$$P_{\mathcal{M}_t} X_{t+h} = V_{t+h} + \sum_{j=h}^{\infty} \psi_j \varepsilon_{t+h-j}, \qquad v_\infty(h) = \sigma_\varepsilon^2 \sum_{j=0}^{h-1} \psi_j^2.$$

The deterministic part contributes no prediction error. As $h \to \infty$, the error variance tends to $\mathrm{Var}(U_t)$.

Theorem 5.7.3 (Spectral decomposition; Theorem 5.7.2)**.** Under $\sigma_\varepsilon^2 > 0$, the spectral measure decomposes as $F_X = F_U + F_V$, where F_U is absolutely continuous and F_V is singular. The density of U is

$$f_U(\lambda) = \frac{\sigma_\varepsilon^2}{2\pi} \left| \sum_{j=0}^{\infty} \psi_j e^{-ij\lambda} \right|^2 .$$

The Fourier series is interpreted as an L^2 boundary function. The assumption $\sigma_\varepsilon^2 > 0$ is essential: a deterministic process may have an absolutely continuous spectral measure.

5.8 Kolmogorov's Formula (Section 5.8)

Theorem 5.8.1 (Kolmogorov; Theorem 5.8.1)**.** Let f be the density of the absolutely continuous part of the spectral measure of a real centered stationary process. Its infinite-past one-step prediction error variance is

$$\boxed{\sigma_\varepsilon^2 = 2\pi \exp\left\{ \frac{1}{2\pi} \int_{-\pi}^{\pi} \log f(\lambda)\, d\lambda \right\}.}$$

The convention is $\exp(-\infty) = 0$. Consequently,

$$\sigma_\varepsilon^2 > 0 \quad \Longleftrightarrow \quad \int_{-\pi}^{\pi} \log f(\lambda)\, d\lambda > -\infty.$$

The singular part of the spectral measure does not enter this formula.

Corollary 5.8.2 (Pure nondeterminism in spectral terms)**.** A nonzero stationary process is purely nondeterministic precisely when its spectral measure is absolutely continuous and its density satisfies $\int_{-\pi}^{\pi} \log f(\lambda)\, d\lambda > -\infty$. Positivity of f almost everywhere alone is insufficient; logarithmic integrability is also required.

A

A.1

Theorem A.1.1 (Proof for the Spectral Thoerem)**.**

Proof. First notice that

$$\|T\| = \sup_{\|x\|\leq 1} |\langle Tx, x\rangle|,$$

for the proof, let us denote the supremum on the right side to be M, then we know for $x, y \in H$, we have

$$\langle T(x+y), x+y\rangle = \langle Tx, x\rangle + \langle Ty, y\rangle + 2\mathrm{Re}(\langle Tx, y\rangle)$$

and hence

$$\langle T(x+y), x+y\rangle - \langle T(x-y), x-y\rangle = 4\mathrm{Re}(\langle Tx, y\rangle)$$

which gives

$$|4\mathrm{Re}(\langle Tx, y\rangle)| \leq M(\|x+y\|^2 + \|x-y\|^2).$$

We will have

$$4\|Tx\| \leq 4M$$

with $y = (Tx)^0$ and $\|x\|| \leq 1$. Therefore, the equality goes and with this, there exists x_n in the unit ball such that

$$\langle Tx_n, x_n\rangle \to M$$

then we know there is a cumulation x of $\{\pm x_n\}$ such that

$$\langle Tx, x\rangle = M \implies \|Tx - Mx\|^2 \leq 2M^2 - 2M^2 = 0$$

and hence $Tx = Mx$ which means there is a non-zero eigenvector of T. It is easy to check that every eigenvalue is real and eigenvector with distinct eigenvalue is orthogonal. To see that an eigenspace is finite-dimensional, assume that $Te_i = \lambda e_i$ for $i \in I$ for a infinite index set I. We may construct an orthonormal sequence of e_i and we know $\|Te_i - Te_j\| = \sqrt{2}|\lambda|$ which is a contradiction to T compact, so an eigenspace has to be finite-dimensional. Similarly, for any $\delta > 0$, the eigenvalues of T larger than δ has to be finite and hence they are at most countable with only 0 possible accumulation point.

Now we consider $M := \overline{\mathrm{span}\{e, Te = \lambda e, \lambda \neq 0\}}$ which is a close subspace of H with $H = M^\perp \oplus M$. Then for any $x \in M^\perp$, we know $Tx \in M^\perp$ and hence $T|_{M^\perp}$ is still adjoint compact and hence it has an eigenvector with non-zero eigenvalue if $\|T|_{M^\perp}\| > 0$, which is a contradiction. Therefore, $M^\perp \subset \ker T$, and easy to check the opposite direction and we are done. □